/research

Research system: /research create [–mode tiered|supervisor|competitive] … | list | open | search | show | delete | archive | cluster

Overview

`/research` drives the structured research pipeline: web search plus local file cross-referencing, synthesis, and a self-contained dated report output (`RESEARCH-<YYYY-MM-DD>.md`, with the legacy undated `RESEARCH.md` still read). Projects are named, persisted, and can be listed, resumed, clustered, exported, and imported. Create accepts a rich flag set covering depth, tier, mode, per-stage models, concurrency, timeouts, and source limits. The TUI runs the pipeline on a background task while the status bar shows `[wait] research: <name>....`

Syntax

```
/research help
/research create <name> <topic> [flags]
/research list [--all] [--json]
/research show <name> [--json]
/research open <name>
/research search <query> [--json]
/research delete <name> [--yes]
/research archive <name>
/research resume <name>
/research continue <name> [--message <text>]
/research update <name>
/research cluster <name> [--force]
/research deck <name> [--from <date>] [--to <date>] [--elements <list>] [--include-old] [--out <p
/research export <name> [--output <path>]
/research import <path> [--name <name>]
/research config
```

The first positional after `create` is the project name; everything after it is the topic. The parser splits on whitespace and honours double quotes, so a topic can be quoted as one argument.

Report title

When the synthesis came from the model, the report's `title` (the frontmatter field and the `# Title:` header) is named by the LLM from the generated executive summary, seeded with the research topic, and capped at 80 characters on a word boundary. A model-produced title reads as a label for the report instead of a clipped first sentence. If the model returns nothing usable, the call fails, or the synthesis fell back to the mechanical path, the title stays the topic-derived one (or the truncated summary) and the run continues. The step reports as a `title` `RunStep` event.

Subcommands

Form	Description
`create <name> <topic>`	Starts a research project. See the flag table below.
`list [--all] [--json]`	Lists projects; `--all` includes archived, `--json` emits JSON.
`show <name> [--json]`	Shows project state and findings; `--json` emits JSON.
`open <name>`	Opens the project's resolved report (newest dated `RESEARCH-<date>.md`, or the legacy `RESEARCH.md`) in the log panel.
`search <query> [--json]`	Searches stored research content.
`delete <name> [--yes]`	Deletes a project; `--yes` skips confirmation.
`archive <name>`	Moves a project to the archive.
`resume <name>`	Resumes an interrupted or archived project.
`continue <name> [--message <text>]`	Continues an existing project with an optional steering message.
`update <name>`	Refreshes an existing project's findings.
`cluster <name> [--force]`	Runs research clustering on the project; `--force` re-runs over existing clusters.
`deck <name> [flags]`	Writes a Markdown diff deck comparing two of the item's dated reports (default: the two newest) to `DECK-<YYYY-MM-DD>.md` in the item directory. See the deck flag table below and research.md for the workflow.
`export <name> [--output <path>]`	Exports the project bundle to a path.
`import <path> [--name <name>]`	Imports a project bundle, optionally renaming it.
`config`	Shows the resolved research configuration.
`help`	Prints the subcommand table.

Create flags

Flag	Values / Default
`--from-url`, `--from-url` (repeatable)	Seed the run from one or more URLs.
`--from-file`, `--from-file` (repeatable)	Seed the run from local files.
`--iterations N`	Iteration budget for the gather loop.
`--depth`	`shallow` | `standard` | `deep`.
`--tier`	`light` | `full` | `dissertation`.
`--mode`	`tiered` | `supervisor` | `competitive`. `competitive` implies `--format comparison-table`.
`--summarization-model`	`<provider:model>` override for summarisation.
`--research-model`	`<provider:model>` override for the researcher stage.
`--compression-model`	`<provider:model>` override for compression.
`--final-report-model`	`<provider:model>` override for the final report.
`--max-concurrent-research-units N`	Parallel research unit cap.

Flag	Values / Default
`--format`	`report` | `executive-summary` | `comparison-table` | `source-bibliography` | `imrad` | `newsletter`.
`--sources-dir <dir>`	Local sources directory to cross-reference.
`--template <name>`	Output template override.
`--fetch-concurrently N`	Concurrent fetch cap (default 10).
`--local-concurrently N`	Concurrent local-file reads (default 8).
`--fetch-timeout-secs N`	Per-fetch timeout (default 30).
`--web-phase-timeout-secs` / `--web-time N`	Web phase budget.
`--local-phase-timeout-secs N`	Local phase budget.
`--search-max-retries N`	Search retry count (default 2).
`--search-retry-base-delay-ms N`	Retry backoff base (default 200).
`--max-web-results N`	Web result cap.
`--max-search-calls N`	Search-call cap.
`--max-queries N`	Cap on the queries produced by topic decomposition (default 20; `0` = unbounded). Per-run override of `research.max_queries`.
`--max-local-sources N`	Local source cap.
`--max-synthesis-sources N`	Sources admitted into synthesis.
`--max-concepts N`	Output cap on the `## Concepts` list (default 5; `0` = unbounded). Per-run override of `research.max_concepts`.
`--max-findings N`	Output cap on the `## Findings` list (default 20; `0` = unbounded). Per-run override of `research.max_findings`.
`--brief <text>`	Provide the brief up front, skipping clarification.
`--use-local`	Include local file sources.
`--use-specs`	Include spec documents as sources.
`--use-low-relevance`	Keep low-relevance hits.
`--papers`	Include scholarly-paper engines (OpenAlex) in the search sweep. Excluded by default.
`--use-pdf`	Enable PDF extraction.
`--clarify` / `--no-clarify`	Force on or off the clarification stage. Off by default.
`--evaluate`	Enable evaluation scoring of findings.
`--url-cloak`	Emit web source URLs as defanged plain text (`hxxps://host[.]tld/...` in a code span) in the `Sources` bullets and `References Index` table. Off by default.
`--age <span>`	Reject web sources published longer ago than the span (`30d`, `6m`, `2y`); undated sources are rejected too. The window's start date (`today - span`, as ISO `YYYY-MM-DD`) is appended to every decomposed search query so the engines steer towards material inside the window.

Deck flags

Flag	Values / Default
`--from <YYYY-MM-DD>`	Older (baseline) report; alone, pairs with the next newer report.
`--to <YYYY-MM-DD>`	Newer (current) report; alone, pairs with the next older report.
`--elements <list>`	Comma-separated subset of `Executive Summary`, `Concepts`, `Findings`, `Cross-References`, `References` (case-insensitive). Default: all.
`--include-old`	Also emit a slide for an element whose items are all unchanged.
`--out <path>`	Write the deck here instead of `DECK-<YYYY-MM-DD>.md` in the item directory.
`--force`	Overwrite an existing deck file.

Examples

```
/research create rust-async "Compare tokio and async-std task scheduling models"
```

Names the project `rust-async`, uses everything after it as the topic.

```
/research create deep-dive "tree-sitter incremental parsing" --tier dissertation --depth deep
```

Runs a deep, dissertation-tier project.

```
/research create db-compare "sqlite vs duckdb analytics" --mode competitive
```

Competitive mode forces `--format comparison-table` output.

```
/research create local-audit "unsafe usage" --sources-dir ./crates --use-local --use-specs
```

Cross-references local crates and spec documents.

```
/research create fast "tokio broadcast channels" --tier light --fetch-concurrently 20 --web-phase
```

A light run with a raised fetch concurrency and web budget.

```
/research create brief "vector databases" --max-concepts 3 --max-findings 10
```

Caps the report to 3 concepts and 10 findings (most-relevant-first).

```
/research create exhaustive "multi-agent orchestration" --max-concepts 0 --max-findings 0
```

Disables both output caps (`0` means unbounded).

```
/research create vendors "managed vector database vendors" --url-cloak
```

Defangs web URLs in the report so automated scanners do not treat them as clickable links.

```
/research create rerun "existing project" --summarization-model ollama:qwen2.5:1.5b
```

Routes the summarisation stage to a fast local model.

```
/research list --all
```

Lists every project including archived ones.

```
/research continue rust-async --message "Focus on the cancellation paths"
```

Continues a project with steering guidance.

Output

- While running, the status bar shows `[wait] research: <name>...` and the pipeline streams progress lines into the message window.
- On completion the project's `RESEARCH-<YYYY-MM-DD>.md` report is written and a summary block lists the report path, source counts, and synthesis highlights.
- `list` renders a project table; `--json` variants emit machine-readable output for scripting.
- Failures print the failing stage with a corrective message; the project remains resumable via `/research resume`.

Related

- `ragent research` CLI runs the same engine outside the TUI.
- `GET/POST/DELETE /research` HTTP endpoints mirror these operations.
- `/research cluster` relies on the research clustering engine.
- See docs/howtos/research.md for the research system internals.